28-2a Stalemate, 1914–1917

The war that erupted in 1914 acquired the name the "Great War" until the 1940s, when a far greater one overshadowed it. Its form came as a surprise to all belligerents, from the generals on down. In the classic battles that every officer studied, the advantage always went to the fastest-moving army led by the boldest general. But in 1914, the generals' carefully drawn plans went awry from the start. Believing that a spirited attack would always prevail, French generals hurled their troops, dressed in bright blue-and-red uniforms, against the well-defended German border and suffered a crushing defeat. In battle after battle, the much larger German armies defeated the French and the British. By early September, the Germans held Belgium and northern France with Paris just over the horizon.

German victory seemed assured. But German troops, who had marched and fought for a month, had reached a point of exhaustion, and their generals wavered. A gap opened between two German armies along the Marne River, into which General Joseph Joffre moved France's last reserves. The Battle of the Marne (September 5–12, 1914) threw the Germans back several miles.

During the next month, both sides spread out until they formed an unbroken line extending over 300 miles (some 500 kilometers) from the North Sea to the border of Switzerland. All along this **Western Front**, the opposing troops prepared their defenses. Machine guns, their most potent weapons, provided an almost impenetrable defense against advancing infantry, yet could not be used for the offensive because their weight required more than one man to carry and they took too much time to set up.

To escape the deadly streams of bullets, soldiers dug holes in the ground, connected the holes to form shallow trenches, then dug communications trenches to the rear. Within weeks, lines of trenches several feet deep, their tops protected by sandbags and their floors covered with planks, crisscrossed the battlefields. Though not entirely new, the trenches along the entire Western Front interconnected so tightly as to leave no gaps through which armies could advance (see Map 28.1). How, then, could either side ever hope to win?

For four years, generals on each side again and again ordered their troops to attack. In battle after battle, thousands of young men on one side climbed out of their trenches and raced across open fields, only to be mown down by enemy machine-gun fire. Hoping to destroy the machine guns, the attacking force would saturate the entrenched enemy lines with artillery barrages. But this tactic alerted the defenders to an impending attack and allowed them to rush in reinforcements and set up new machine guns.

The year 1916 saw the bloodiest and most futile battles of the war. The Germans attacked French forts at Verdun, losing 281,000 men and causing 315,000 French casualties. In retaliation, the British attacked the Germans at the Somme River and suffered 420,000 casualties—60,000 on the first day alone—while the Germans lost 450,000 and the French 200,000.

Warfare had never been waged this way before: mass slaughter in a moonscape of mud, steel, and flesh. Both sides attacked and defended, but neither side could win, for the armies were stalemated by trenches and machine guns. During four years of the bloodiest fighting the world had ever seen, the Western Front moved no more than a few miles one way or another.

At sea, the conflict proved just as inconclusive. As soon as war broke out, the British cut the German overseas telegraph cables, blockaded the coasts of Germany and Austria–Hungary, and set out to capture or sink all enemy ships still at sea. The German High Seas Fleet, built at enormous cost but mostly coal-fueled, seldom left port. Only once, in May 1916, did it confront the oil-fueled and thus more efficient British Grand Fleet. At the Battle of Jutland, off the coast of Denmark, the two fleets lost roughly equal numbers of ships before the Germans escaped back to their harbors.

In early 1915, in retaliation for the British naval blockade, Germany announced a blockade of Britain by submarines. German submarines attacked every vessel they could. One of their victims, the British ocean liner *Lusitania,* took the lives of 1,198 people, 139 of them Americans. When the United States protested, Germany ceased its submarine campaign, hoping to keep America neutral.

Airplanes found use in reconnaissance and engaged in spectacular but inconsequential dogfights above the trenches. They also dropped bombs that terrified civilian populations., Poison gas, introduced on the Western Front in 1915, killed and wounded attacking soldiers as well as their intended victims, adding to the horror of battle. Primitive tanks aided, but did not cause, the collapse of the German army in the last weeks of the war. Although these weapons did not have great strategic impact on World War I, they offered a preview of future warfare.

Western Front A line of trenches and fortifications in World War I that stretched without a break from Switzerland to the North Sea. Scene of most of the fighting between Germany, on the one hand, and France and Britain, on the other.

28-2b The Home Front and the War Economy

Trench-bound armies demanded ever more weapons, ammunition, and food, so civilians had to work harder, eat less, and pay higher taxes. Textiles, coal, meat, fats, and imported products such as tea and sugar became strictly rationed. Governments gradually imposed stringent controls over all aspects of their economies.

The war economy transformed civilian life. In France and Britain, food rations allocated according to need actually improved nutrition among the poor. Unemployment vanished. Thousands of Africans, Indians, and Chinese entered Europe to supply heavy labor. Employers hired women to fill jobs vacated by men off to war. Some women became streetcar drivers, mail carriers, and police officers. Others found work in the burgeoning government bureaucracies. Many served as doctors, nurses, mechanics, and ambulance drivers in auxiliary military services for the British government after 1917, as the war took its heavy toll of young men. These positions gave thousands of women a sense of participation in the war effort and a taste of personal and financial independence.

German civilians paid an especially high price for the war, for the British naval blockade severed their overseas trade links. Wheat flour disappeared, replaced first by rye, then by potatoes and turnips, then by acorns and chestnuts, and finally by sawdust. After the failure of the potato crop in 1916 came the "turnip winter," when people had to survive on 1,000 calories per day, half the normal amount that an active adult requires. Shortages especially affected women, children, and the elderly. Soldiers at the front raided enemy lines to scavenge food.

The war also brought hardships to Europe's African colonies. When the war began, the British and French overran German Togo on the West African coast. The much larger German colonies of Southwest Africa and German Cameroon fell in 1915. In German East Africa, the Germans remained undefeated until the end of the war. The Europeans requisitioned foodstuffs, imposed heavy taxes, and forced Africans to grow export crops and sell them at low prices. As Europeans stationed in Africa joined the war, the combination of increased demands on Africans and fewer European colonial officials led to uprisings that lasted for several years. Over a million Africans served in the various armies, perhaps three times that number being drafted as porters to carry army equipment. Faced with a shortage of young Frenchmen, France drafted Africans into its army, where many fought side by side with Europeans.

One country grew rich during the war: the United States. For two and a half years the United States stayed technically neutral but did a roaring business supplying France and Britain. When the United States entered the war in 1917, businesses engaged in war production made spectacular profits. Responding to government urging, civilians helped the war effort by investing their savings in war bonds and growing food in backyard "victory gardens." Employment opportunities created by the war played a major role in the migration of African Americans from the rural South to the cities of the North.

Women in World War I Women played e in World War I than in previous wars. d millions of men, employers hired women k. This poster extolls the importance of pplying munitions. Pictorial Press Ltd/Alamy



28-2c The Ottoman Empire at War

On August 2, 1914, the Turks signed a se
In November, they joined t
expense. Dur

Environment & Technology

New War, New Tools

The American Civil War (1860–1865) and the seven-month Franco-Prussian War (1870–1871) had been the biggest conflicts between more or less evenly equipped armies in the era following Napoleon's defeat at Waterloo (1815). The Crimean War (1853–1856), though a major conflict, pitted a Russian army that had seen few improvements since Napoleon's time against more advanced French and British forces supporting the Ottoman Empire. The Opium Wars, Arrow War, and Taiping Rebellion in China; the Zulu War in South Africa; and other colonial conflicts featured even more lopsided terms of equipment.

World War I introduced a new type of warfare: larger artillery pieces firing heavier shells softened up fixed entrenchments from long distance prior to mass attacks across the no-man's-land between trench lines. Infantrymen armed primarily with rifles and grenades charged into withering defensive fire from machine guns. Mounted on tanks starting in 1916, machine guns and cannon became mobile offensive weapons, though rough terrain caused many tanks to break down. At sea, the ironclad warships from the American Civil War grew into mammoth dreadnoughts equipped with immense guns. In the air, balloons used for gaining aerial views of battlefield movements shared the sky with primitive warplanes and, occasionally, Zeppelins carrying bombs. All of these developments followed clear nineteenth-century precedents.

Other innovations went beyond guns and explosives and drew on other spheres of endeavor. Germany's chemists, the best in the world, weaponized poison and disabling gases. Ignoring an international Hague Declaration against "asphyxiating gases" (1899) and a Hague Convention on Land Warfare (1907) that forbade "poison or poisoned weapons," Germany inaugurated gas warfare with greenish chlorine gas in 1915 and added mustard gas to its arsenal in 1917. A French Nobel Prize winner in chemistry responded with colorless and more difficult to detect phosgene. Britain did not launch its own gas attacks until 1918, and the United States had begun production of its own lethal invention, lewisite, but had not deployed it in battle before the war ended in 1918.

Germany's chemists also developed flamethrowers, though their range of twenty yards and limited fuel supply restricted their usefulness. Britain experimented with larger flame projectors but did not find a practical use for them.

Besides weapons and vehicles, including lifesaving ambulances, heavy industry provided miles of concertina wire, cylindrical rolls of barbed wire strung between upright poles that became a standard obstacle in the no-man's-land between trenches. Germany produced Dannert wire, a high-grade, tempered steel version that proved hard to cut and did not need poles.

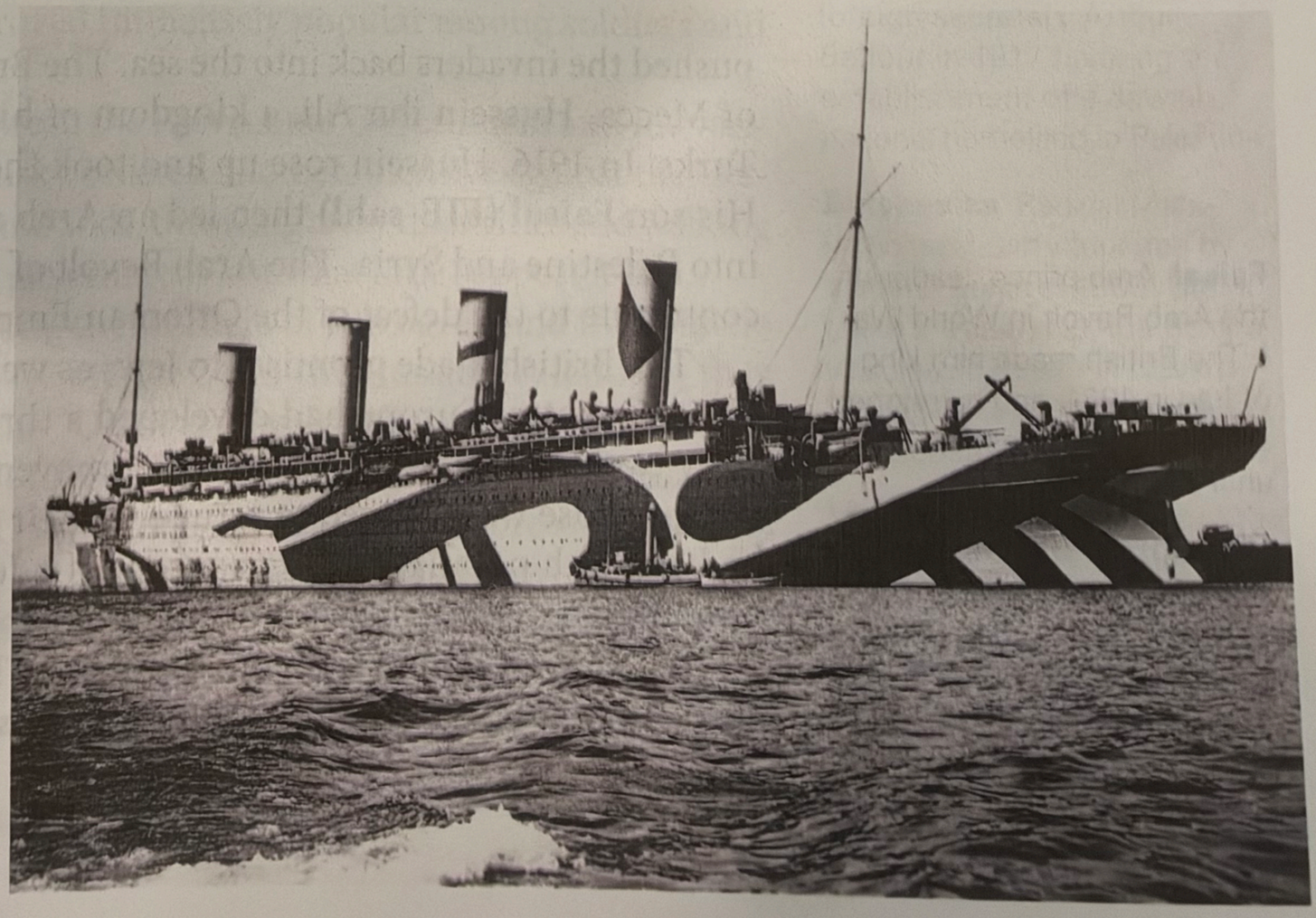
Field telephones improved battlefield communications immensely, though unreeling wire could be a dangerous assignment. Radio began at a more primitive stage but developed rapidly during the war, particularly for naval use. Field sets depended on fragile vacuum tubes, and visible aerials became targets for enemy fire.

As for diet, the "iron ration" carried into the field by British soldiers consisted of 1 lb. canned corned beef or Maconochie, a barely edible stew; 12 oz. biscuit; 5/8 oz. tea; 2 oz. sugar; 1/2 oz. salt; 3 oz. cheese; and 1 oz. meat extract (2 cubes.) The German equivalent gave each soldier 250g (8.8 oz) biscuit; 200g (7 oz.) preserved meat or 170g (6 oz.) bacon; 150g (5.3 oz.) preserved vegetables; 25g (9/10 oz.) coffee; and 25g (9/10 oz.) salt.

Even the art world contributed new tools of war. Artist Lucien-Victor Guirand de Scévola, the leader of the French camouflage (from Parisian slang *camoufler*, meaning "to disguise") corps, sought to apply cubist principles by using paint and other materials to break up easily identifiable battlefield outlines. One artist, for example, disguised an observation post to look like a tree. British artist Norman Wilkinson came up with "dazzle camouflage," which consisted of painting ships with radically garish geometric patterns that made it difficult for German submarine commanders to tell the speed and direction of ships in convoys.

Questions for Analysis

1. How did changes in warfare reflect economic developments in the late nineteenth century?
2. Why might poison gas have been used despite international prohibitions on its use?
3. Did technology slow down or speed up the pace of war?



▶ **Image 28.3 Dazzle Camouflage** Ocean liner painted with dazzle camouflage being used for Atlantic Ocean convoy activities during World War I. The purpose of the camouflage was to confuse submarine commanders looking through a periscope as to the identity, size, speed, and direction of a surface ship. UK: HMT Olympic in dazzle camouflage while in service as a troopship during World War I/Pictures From History/Bridgeman Images